Probability and Statistics Notes

Prob Stats 1: Notes, Exercises and PYQs

Eklavya Raman
er24ms024@iiserkol.ac.in

January 9, 2026

Contents

List of Theorems

Part I

Probability

Chapter 1

Continous Random Variables and Distributions

1.1 Continous Random Variables

Let us first define what a continous function and a continous random variable is.

Definition 1.1.0.1

Continous Function
A function $f : \mathbb{R} \to \mathbb{R}$ is said to be continous at a point $x_0 \in \mathbb{R}$ if for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all $x \in \mathbb{R}$, if $|x - x_0| < \delta$, then $|f(x) - f(x_0)| < \epsilon$. If a function is continous at every point in its domain, it is called a continous function.

Let us consider a simple example of a continous function ($f(x) = \sin(x)$) -

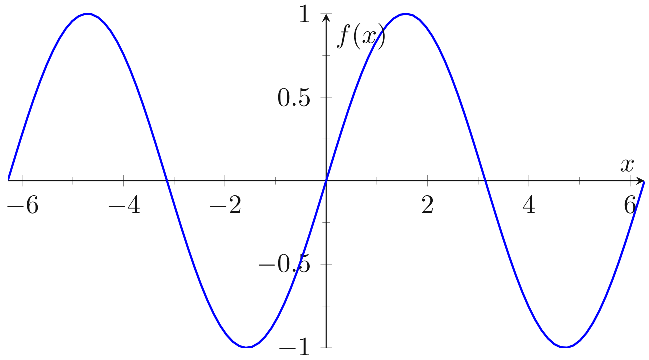

Figure 1.1: Graph of the continous function $f(x) = \sin(x)$

Definition 1.1.0.2

Continous Random Variable
A random variable X is said to be continous if its cumulative distribution function (CDF) $F_X(x) = P(X \leq x)$ is continous for all $x \in \mathbb{R}$.

Now, we define the probability density function (PDF) for a continous random variable.

Definition 1.1.0.3

Probability Density Function (PDF)
The probability density function (PDF) of a continous random variable X is a function $f_X(x)$

such that for any two numbers a and b with $a < b$,

$$P(a < X < b) = \int_a^b f_X(x)\,dx$$

The PDF must satisfy the following properties:

1. $f_X(x) \geq 0$ for all $x \in \mathbb{R}$
2. $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$

We also now define the cumulative distribution function (CDF) for a continous random variable.

Definition 1.1.0.4

Cumulative Distribution Function (CDF)
The cumulative distribution function (CDF) of a continous random variable X is a function $F_X(x)$ defined as

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(t)\,dt$$

where $f_X(t)$ is the probability density function (PDF) of X.

Now, we are equipped to study these continous random variables and their distributions in detail.

Let $X \sim f(x) \wedge F(x)$, let $Y = g(X)$, where g is strictly monotonic and differentiable. Then,

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y) = \begin{cases} P(X \leq g^{-1}(y)), & \text{if g is increasing} \\ P(X \geq g^{-1}(y)), & \text{if g is decreasing} \end{cases}$$

We also have -

$$f_Y(y) = \begin{cases} f(g^{-1}(y)) \cdot \dfrac{d}{dy} g^{-1}(y), & \text{if g is increasing} \\ f(g^{-1}(y)) \cdot \left| \dfrac{d}{dy} g^{-1}(y) \right|, & \text{if g is decreasing} \end{cases}$$

Therefore, we can simply write -

$$f_Y(y) = f(g^{-1}(y)) \cdot \left| \frac{d}{dy} g^{-1}(y) \right| \sim f(x) \cdot |\frac{dx}{dy}|$$

Now, let us consider the case where g is monotone, but not strictly monotone. This means that there exists an interval where g is constant. Let us denote this interval as $[a, b]$. In this case, for any y in the range of g corresponding to the interval $[a, b]$, we have -

$$P(Y = y) = P(g(X) = y) = P(X \in [a, b]) = \int_a^b f_X(x)\,dx$$

Let $g^{-1}(y) = \inf\{x : g(x \geq y)\}$, then we have -

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

Differentiating both sides with respect to y, we get -

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

However, since g is not strictly monotone, we need to account for the intervals where g is constant. Therefore, we can write the PDF of Y as -

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{d}{dy} g^{-1}(y) \right| + \sum_i P(X \in [a_i, b_i]) \cdot \delta(y - g(c_i))$$

where $[a_i, b_i]$ are the intervals where g is constant, c_i is any point in the interval $[a_i, b_i]$, and δ is the Dirac delta function. This accounts for the probability mass at the points where g is constant.

Example

Let $X \sim f(x) = \begin{cases} \frac{1}{2}, & -0.5 \leq x \leq 1.5 \\ 0, & \text{otherwise} \end{cases}$ and $Y = X^2$. We need to find the distribution of Y.

We have -

$$F_Y(y) = P(Y \leq y) = P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y})$$

Therefore,

$$F_Y(y) = \int_{-\sqrt{y}}^{\sqrt{y}} f_X(x)\, dx = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2}\, dx = \frac{\sqrt{y} - (-\sqrt{y})}{2} = \sqrt{y}$$

for $0 \leq y \leq 2.25$. Differentiating, we get -

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{1}{2\sqrt{y}}$$

for $0 < y \leq 2.25$. Thus, the distribution of Y is given by -

$$f_Y(y) = \begin{cases} \frac{1}{2\sqrt{y}}, & 0 < y \leq 2.25 \\ 0, & \text{otherwise} \end{cases}$$